Every-Iterate Quadratic Amplitudes for Finite Cyclic Rotations

Route-A structural certificate C161: first improvement

Abstract

For the odd cyclic rotation $R_q(x) = x + 1$ with quadratic observable $ax^2 + bx$, we evaluate the complete Birkhoff amplitude at every positive iterate. A gcd gate gives exact vanishing, and completion of the square gives sign, phase, and magnitude otherwise. Over a prime source ring the zero level obeys a discriminant law; for $p \geq 5$, $a = 1, b = 0$, its discriminant is $n^2(1 - n^2)/3$. A finite weighted Koopman unitary supplies an exact same-clock trace expression with a compensating shift. Time-reversal closure is deferred to the final round.

Keywords: cyclic dynamics; quadratic Gauss sum; discriminant; finite Koopman unitary; exact verification.

中文摘要：第一轮改进稿在完整高斯和求值之外加入素数阶零水平判别式。对 $p \geq 5$ 的纯二次子族，判别式为 $n^2(1 - n^2)/3$。有限加权库普曼算子还给出同钟迹表达式，其中补偿平移不可省略。反西时间反演在最终轮次补齐。

关键词：循环动力学；二次高斯和；判别式；有限库普曼算子；精确验证。

1 Amplitude and zero levels

With

$$(A_n, B_n, C_n) = \left(an,\ an(n-1) + bn,\ \frac{an(n-1)(2n-1)}{6} + \frac{bn(n-1)}{2} \right),$$

put $d = (A_n, q)$ and $Q = q/d$. Translation by Q gives exact vanishing when $d \nmid B_n$. Otherwise completing the square gives

$$G_{q,n} = d \left(\frac{A_n/d}{Q} \right) \epsilon_Q \sqrt{Q} \exp\!\left(\frac{2\pi i}{q} \{C_n + dh\} \right), \quad h = -\{4A_n/d\}^{-1}(B_n/d)^2 \bmod Q,$$

with the separate constant branch when $Q = 1$. For odd prime p, a nonzero quadratic coefficient gives

$$\#\{x : S_n\phi(x) = 0\} = 1 + \left(\frac{B_n^2 - 4A_nC_n}{p} \right).$$

For $p \geq 5$, $a = 1, b = 0$, and $n \not\equiv 0 \bmod p$, the discriminant is $n^2(1 - n^2)/3$; $n \equiv \pm 1$ gives the single double root, while $n \equiv 0$ gives all p roots.

2 Same-clock unitary

On $\ell^2(\mathbb{Z}/q\mathbb{Z})$ let $(Kf)(x) = f(x + 1)$, $M_\phi f = e^{2\pi i\phi/q} f$, and $U = M_\phi K$. Direct iteration yields

$$G_{q,n} = \mathrm{Tr}(U^n K^{-n}).$$

The compensating shift is essential. This is a source finite-unitary identity, not a target trace formula.

Declarations

All claim-bearing data and code are packaged; no external dataset, subjects, or funding is involved. No competing interest is known. An AI coding assistant supported drafting and code review but was not an external reviewer.